

Khaled Lakhdher

Computer Engineer | AI & Data Science | Full-Stack AI Developer

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PROFILE

AI and data engineer who turns models and data into dependable products. I build LLM and RAG systems, multi-agent platforms, and end-to-end ML pipelines, with a metrics-first focus on data quality, evaluation, and clean deployment. Comfortable across the stack: Python, SQL, FastAPI, and Docker.

EXPERIENCE

AI Engineer - Final Year Project (Graduation)

Feb 2026 - June 2026 · Tunisia

CTT - Carthage Training & Technologies

- Designed and developed **Hotel Mind**, a multilingual, multi-agent hotel search and booking assistant: 9 LangChain/LangGraph agents call 32+ tools across 10 real SOAP endpoints for live hotel, room, and pricing data.
- Grounded every answer in live tool output, not model memory, via a hybrid flow (AI search, a guest-detail form, a sandboxed SOAP reservation) on a full-stack FastAPI + Next.js app with LangSmith tracing.

Artificial Intelligence Developer Intern

July 2025 - September 2025 · Hammamet

CTT - Carthage Training & Technologies

- Fine-tuned a Hugging Face NLP model to classify customer reviews into three sentiment classes, improving accuracy by 25% over the baseline.
- Built a Django backend to manage customer data and automate export of analysis results to Excel for reporting.
- Owned the end-to-end Python pipeline for preprocessing, training, and integration.

PROJECTS

Nexus AI - Artificial Intelligence Employee Management Platform

- Engineered a multi-tenant platform (FastAPI, PostgreSQL/pgvector, Redis, Next.js) governing LLM agents with IAM permissions, vector memory, multi-agent orchestration over a Redis message bus, and real-time WebSocket monitoring, behind a provider-agnostic model gateway (Anthropic, OpenRouter, Gemini).

DecisionFlow - AI Business Analyst (Multi-Tenant SaaS)

- Built a multi-tenant analytics SaaS (FastAPI, PostgreSQL/RLS, DuckDB, Polars, Next.js) turning an uploaded CSV into a star schema, dashboard, forecast, and natural-language Q&A, with LLM-generated SQL sandboxed across 3 layers and tenant isolation enforced in-database.

CodeReview Evaluation - Benchmarked LLM Code Reviewer

- Developed an LLM code-review and evaluation framework on the BugsInPy benchmark (500 real bugs); using an LLM-as-judge, showed a single detection score overstates capability by $\sim 1.8\times$ (78% localization vs. 43% diagnosis) across 3 model tiers, with a live Streamlit demo.

CERTIFICATIONS

- Fundamentals of Deep Learning - NVIDIA (Nov 22, 2022)
- Building Transformer-Based Natural Language Processing Applications - NVIDIA (Dec 19, 2025)
- Applications of AI for Predictive Maintenance - NVIDIA (Oct 27, 2025)

SKILLS

Programming Languages: Python · Java · C · SQL · JavaScript · TypeScript

Frontend Technologies: React.js · React Native · HTML/CSS · Tailwind CSS

AI Libraries & Frameworks: Transformers · PyTorch · TensorFlow · Keras · Hugging Face · scikit-learn · Pandas · NumPy · Matplotlib

Development Frameworks & Tools: Gradio · Streamlit · Flask · Django · Zeep · LangSmith · Docker · Postman · Git · GitHub

Concepts & Technologies: Artificial Intelligence (AI) · Machine Learning (ML) · Deep Learning (DL) · Large Language Models (LLMs) · RAG (Retrieval-Augmented Generation) · AI Agents · Agentic AI · Tool Calling · LLM Evaluation · Natural Language Processing (NLP) · Automatic Speech Recognition (ASR) · GANs · Semantic Search · NoSQL Databases · Cloud Computing

EDUCATION

Computer Engineering - AI & Data Science Specialization

2023 - 2026

EPI - International Polytechnic School · Sousse, Tunisia

Preparatory Cycle

2021 - 2023

EPI - Private International Polytechnic School · Sousse, Tunisia

LANGUAGES

French: DELF B2 **English:** TOEIC (Listening & Reading) - 825/990